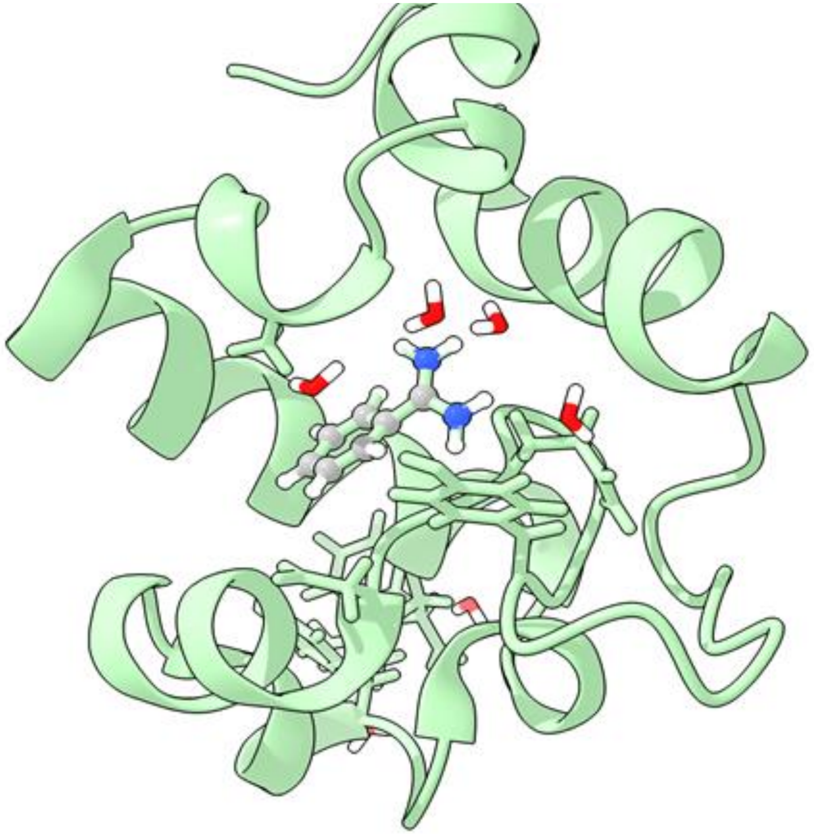
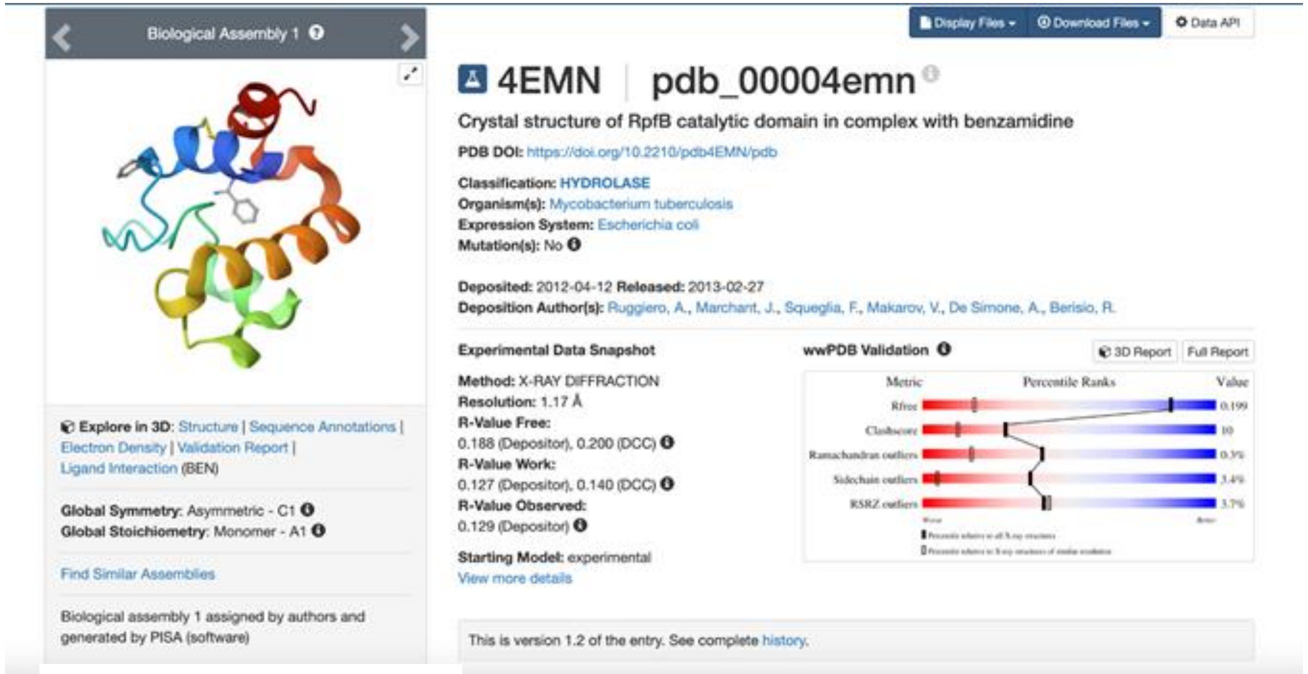
Simulação Proteína-Ligante

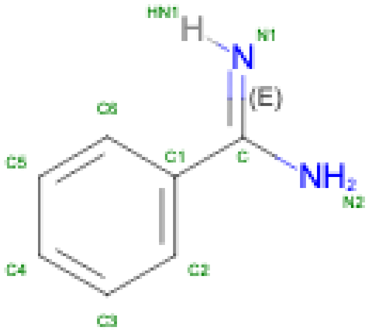


FIOCRUZ SUS

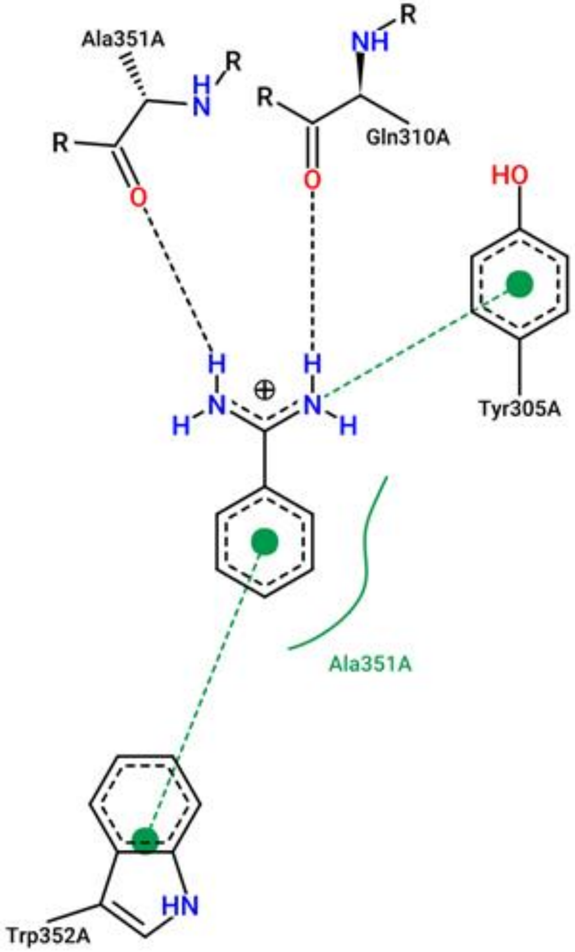
Interação receptor-ligante

Resuscitation promoting factor B from Mycobacterium tuberculosis.



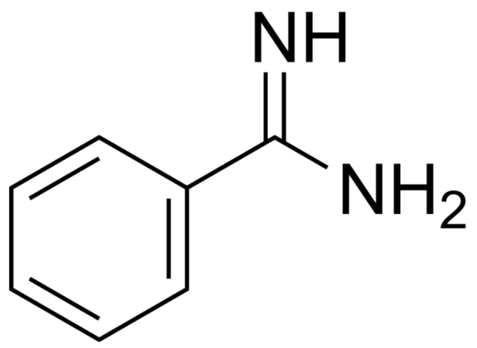


Benzamidina

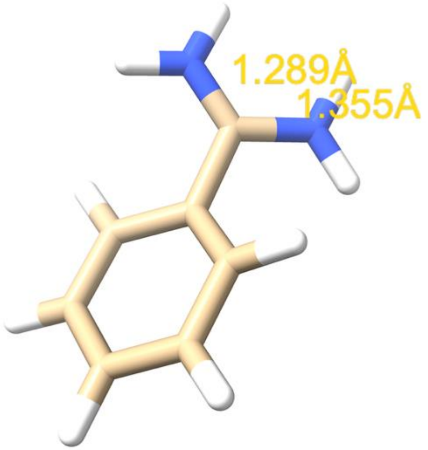


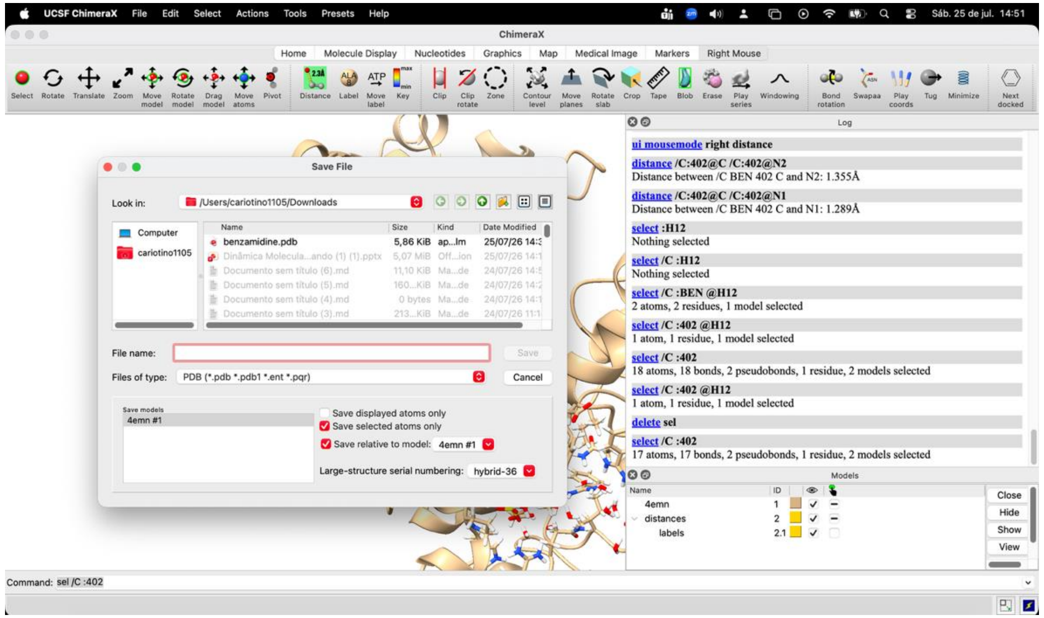
Preparação e parametrização do ligante

- Obter a estrutura usando o programa ChimeraX. (File/Fetch by ID)
- Selecionar um dos ligantes (sel /C :402)
- Adicionar átomos de hidrogênio (addh)
- Corrigir o estado de protonação do C=NH (sel /C :402 @H12)
- Apagar o H que sobra (del sel)
- Selecionar novamente o ligante modificado (sel /C :402)
- Salvar em arquivo (lig.pdb)



Benzamidina





Preparação e parametrização do ligante

b2B ACPYPE Server

Menu

- Home
- API Reference
- Public entries
- Submit
- Status
- Token
- Email

Token: No token

Email: No email

Webserver update on 2025-01-14: Fixing the email notifications

Our team is currently working to reestablish email notification services. In the meantime, we recommend users manually check the results on the status page to stay updated. We appreciate your understanding and patience as we resolve this issue.

Webserver update on 2023-01-24: New version of ACPYPE is now available

We are pleased to announce that our webserver has been updated with the latest version of our service. We apologize for any inconvenience this may have caused and thank you for your patience during this process.

If you have any questions or concerns, please do not hesitate to contact the ACPYPE Server development team (Bio2Byte@vub.be).

About

The ACPYPE Portal was designed to generate topology parameters files for unusual organic chemical compounds. It is based on **ANTECHAMBER** and so far, ACPYPE is able to work for CNS/XPLOR, GROMACS, CHARMM and AMBER.

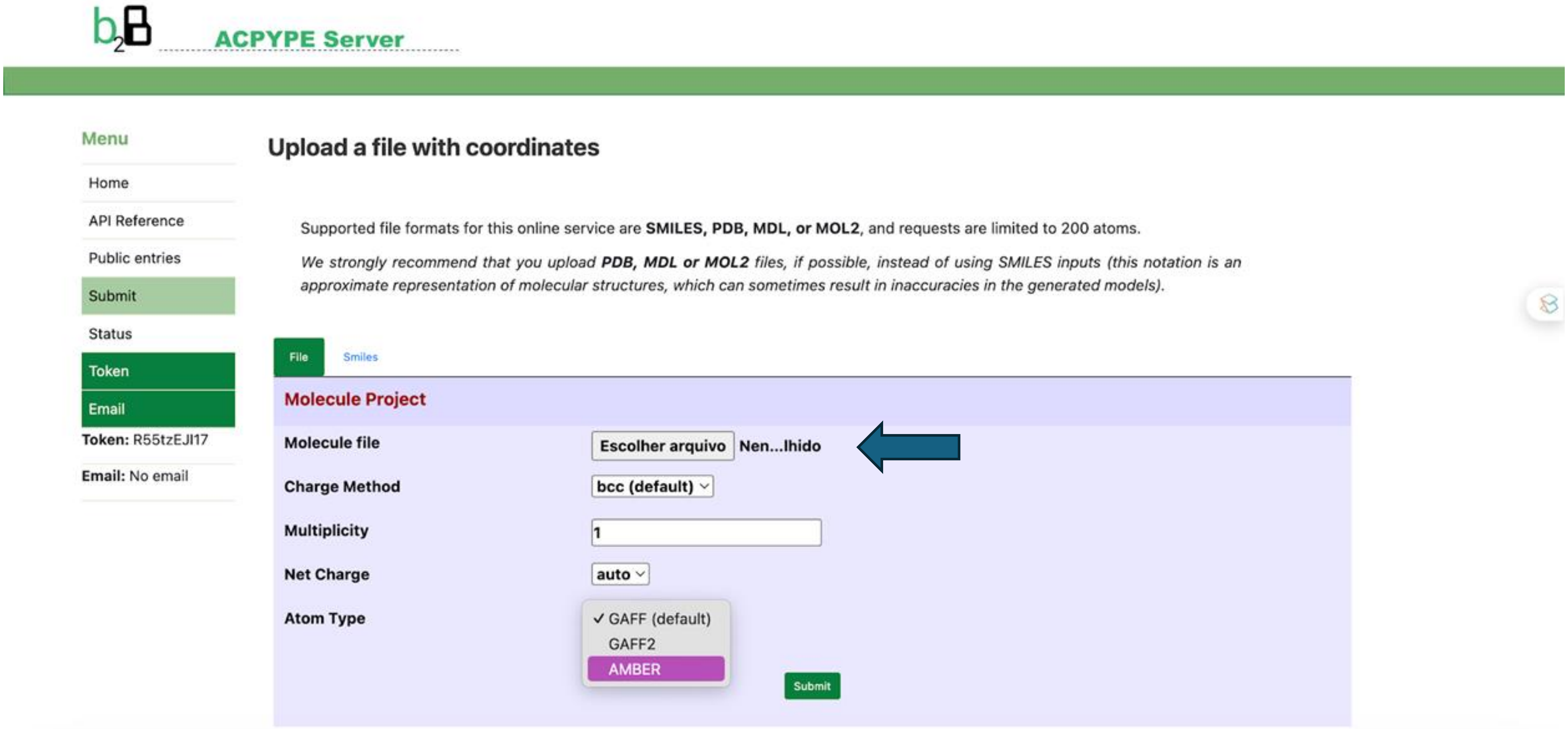
The main scope for ACPYPE Server is to help to pave the way for automatic molecular dynamics simulations involving molecules with unknown parameters, like for example, in complexes of protein and inhibitor, where the ligand is usually an unusual chemical compound.

ACPYPE stands for ***A****nte****C****hamber* ***PY****thon* ***P****arser interfac****E*** and is pronounced "ace + pipe".

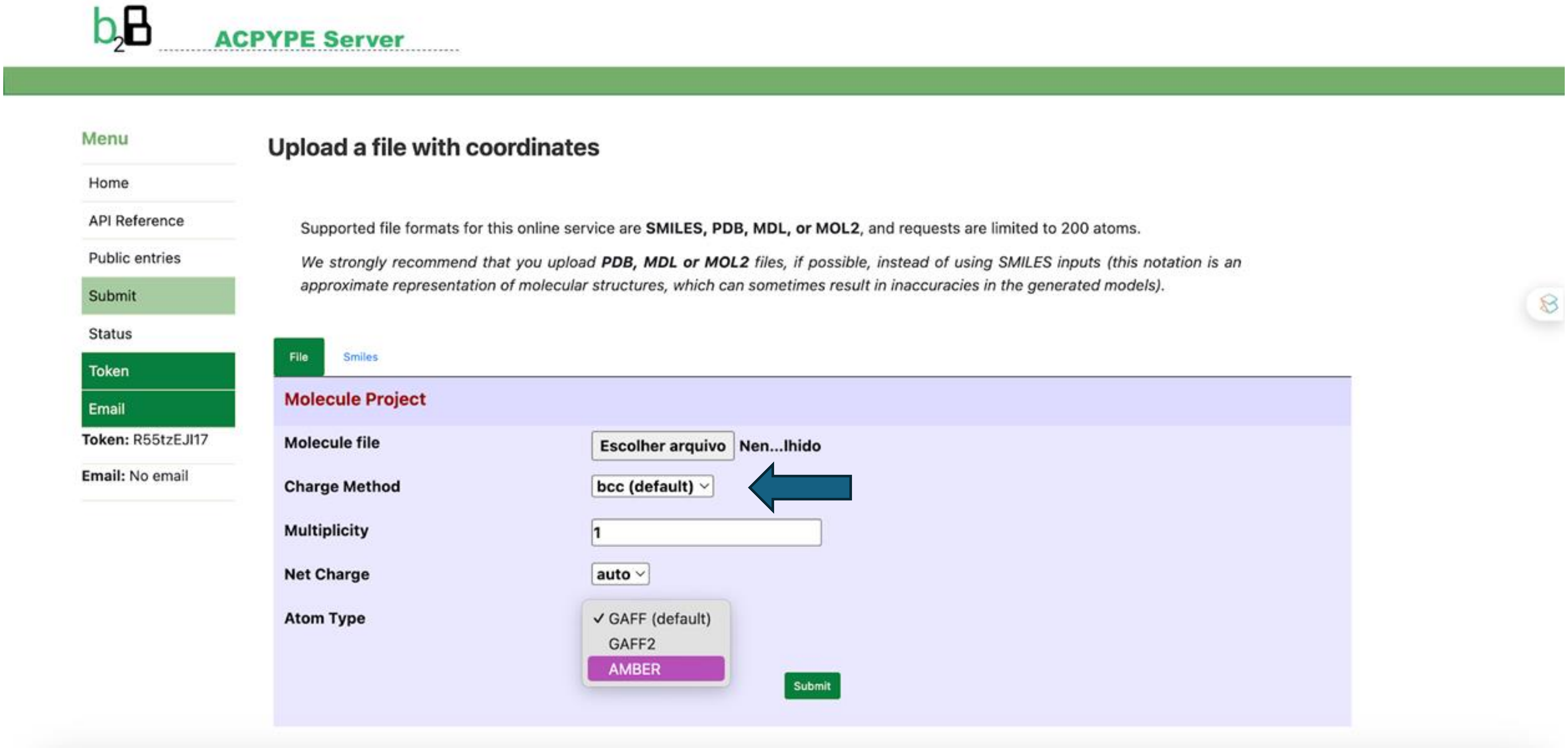
If you use this resource, please cite:

- SOUSA DA SILVA, A. W. & VRANKEN, W. F. ACPYPE - AnteChamber PYthon Parser interfacE. BMC Research Notes 2012, 5:367
- KAGAMI, L. P., SOUSA DA SILVA, A. W., DÍAZ, A., & VRANKEN, W. F., The ACPYPE web server for small-molecule MD topology generation, Bioinformatics, Volume 39, Issue 6, June 2023, btad350

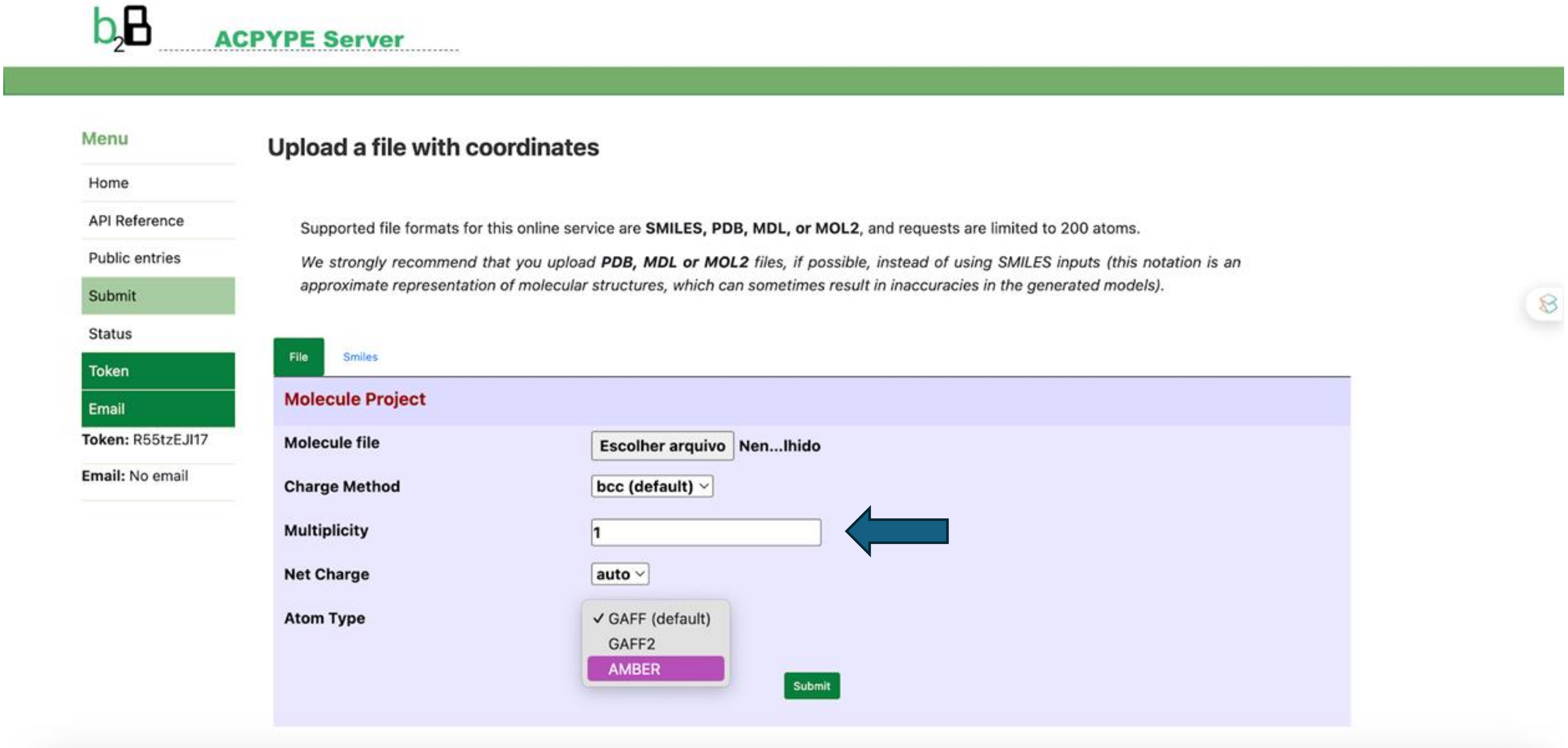
Parametrização do ligante



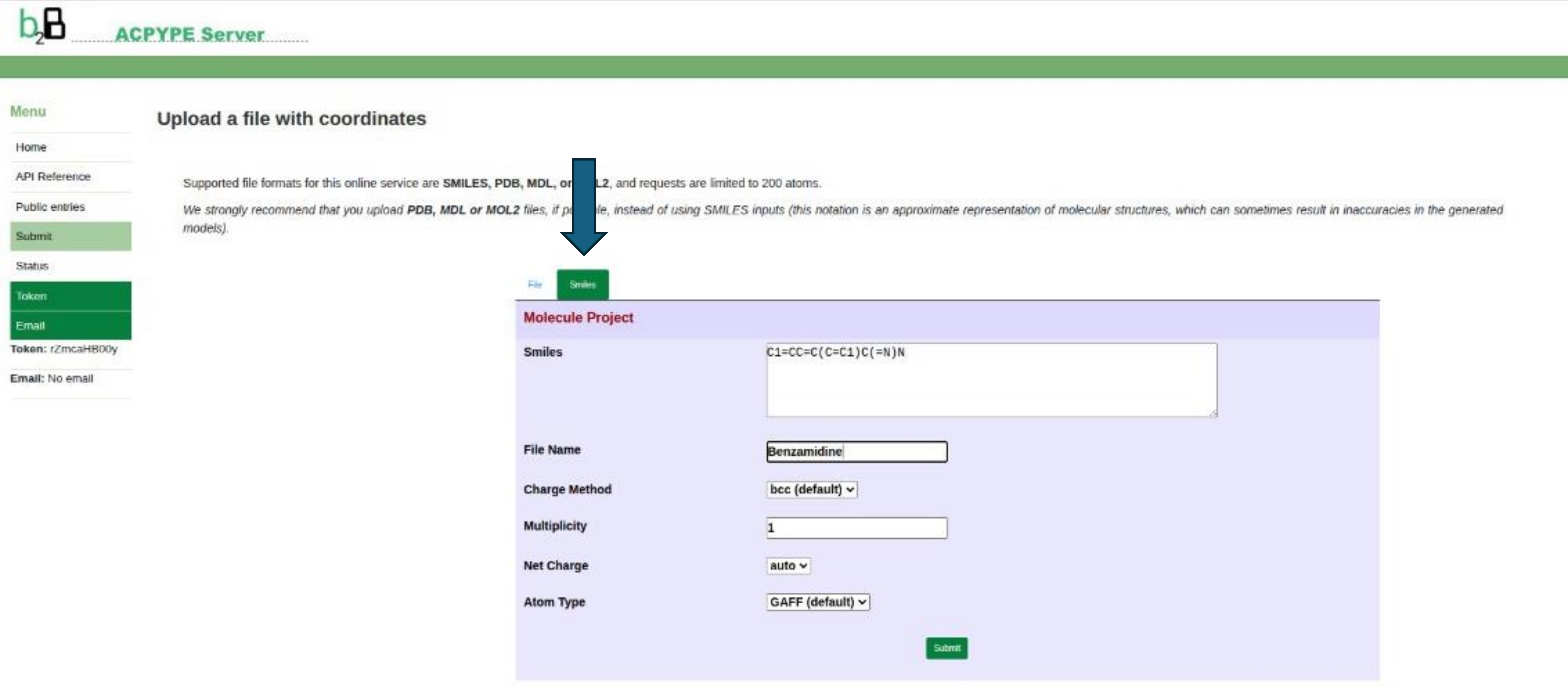
Parametrização do ligante



Parametrização do ligante



Parametrização do ligante – Notação SMILES



Parametrização do ligante – Resultados

ACPYPE Server

Menu

- Home
- API Reference
- Public entries
- Submit
- Status
- Token
- Email

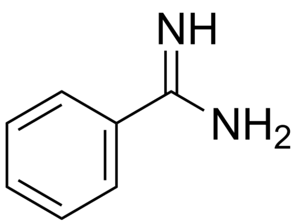
Token: rZmcaHB00y

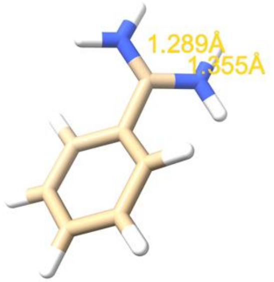
Email: No email

No running jobs.

0 waiting | 0 processing | 0 total

Job	Submitted	Status	Input	Parameters	Identifiers	Actions
Benzamidine	2026-07-29 20:11:36 1 minute ago	Finished	SMILES SMILES: C1=CC=C(C=C1)C(=N)N	**Charge** bcc **Multiplicity** 1 **Net charge** auto **Atom type** gaff	ID: 40974 Hash: bolCXR5cyjvZUvuQQihZ RQ: bolCXR5cyjvZUvuQQihZ	Download Entry API Log Delete
lig.pdb	2026-07-29 20:07:55 4 minutes ago	Finished	INPUT_FILE	**Charge** bcc **Multiplicity** 1 **Net charge** auto **Atom type** gaff	ID: 40973 Hash: H17Ssx1nalHDo9KamMls RQ: H17Ssx1nalHDo9KamMls	Download Entry API Log Delete





Research Foundation Flanders
Opening new horizons

ACPYPE Server Team - 2026

Hosted by the - Wim Vranken, VUB, Belgium

Release version: v20260630.1 (deployed on 2026-06-30) on Tier-1 Cloud (VSC, Belgium)

Verificar se o arquivo de estrutura criado coincide com a estrutura da molécula

Arquivo de topologia – Inclusão do ligante

```
;	File 'topol.top' was generated
;	By user: cariotino1105 (501)
;	On host: MacBook-Air-de-Ernesto.local
;	At date: Tue Mar 25 17:12:02 2025
;
;	This is a standalone topology file
;
;	Created by:
;	                :-) GROMACS - gmx pdb2gmx, 2025.0-Homebrew (-:
;
;	Executable:   /opt/homebrew/bin/../Cellar/gromacs/2025.0/bin/gmx
;	Data prefix:  /opt/homebrew/bin/../Cellar/gromacs/2025.0
;	Working dir:  /Users/cariotino1105/gromacs/curso/dia3
;	Command line:
;	  gmx pdb2gmx -f protein_HOH.pdb -o protein -ter -ignh -ss
;	Force field was read from the standard GROMACS share directory.
;

; Include forcefield parameters
#include "amber99sb-ildn.ff/forcefield.itp”
#include “lig.itp”

[ moleculetype ]
; Name            nrexcl
Protein_chain_C     3

[ atoms ]
;   nr       type  resnr residue  atom   cgnr     charge       mass  typeB    chargeB      massB
; residue 282 GLY rtp NGLY q +1.0
     1         N3    282    GLY      N      1     0.2943      14.01
     2          H    282    GLY     H1      2     0.1642      1.008
     3          H    282    GLY     H2      3     0.1642      1.008
     4          H    282    GLY     H3      4     0.1642      1.008
     5         CT    282    GLY     CA      5      -0.01      12.01
     6         HP    282    GLY    HA1      6     0.0895      1.008
     7         HP    282    GLY    HA2      7     0.0895      1.008
     8          C    282    GLY      C      8     0.6163      12.01
     9          O    282    GLY      O      9    -0.5722         16   ; qtot 1
; residue 283 SER rtp SER  q  0.0
```

Energia de Interação proteína-ligante

Command lines:

gmx grompp -f **md_NPT_rerun.mdp** -o md_NPT_rerun -c md_NPT.gro -p topol.top -maxwarn 3

gmx mdrun -v -s md_NPT_rerun.tpr -deffnm md_NPT_rerun -rerun md_NPT_10ns.trr

Transformar a trajetória em formato xtc (comprimido e só coordenadas)

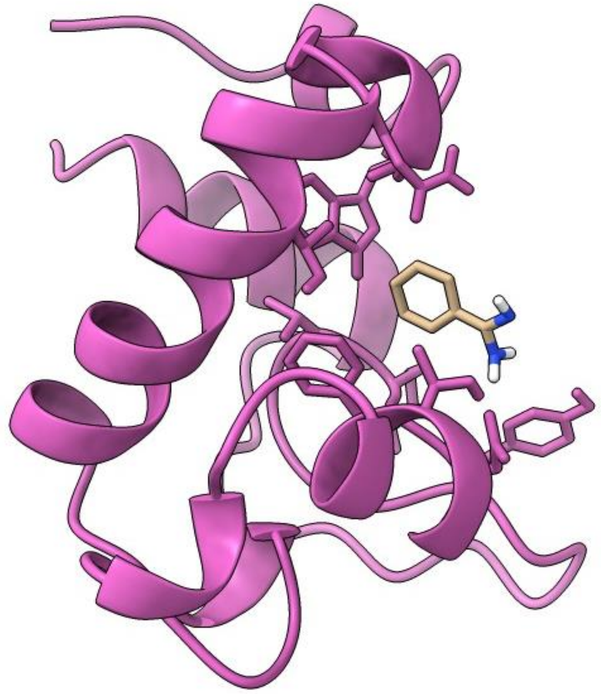
gmx trjconv -s md_NPT_rerun -f md_NPT_10ns.trr -o md_NPT_10ns.xtc -pbc nojump

```
; Pressure coupling is on for NPT
Pcoupl                = Parrinello-Rahman
tau_p                 = 1.0
Compressibility       = 4.5e-05
ref_p                 = 1.0
; Energy groups definition
energygrps            = Protein SOL BEN
; Do not generate velocities
gen_vel               = no
```

Adicionar ao arquivo de parâmetros mdp

Select the terms you want from the following list by
selecting either (part of) the name or the number or a combination.
End your selection with an empty line or a zero.
----

1 Bond 2 Angle	3 Proper-Dih.	4 Per.-Imp.-Dih.	
5 LJ-14 6 Coulomb-14	7 LJ-(SR)	8 Disper.-corr.	
9 Coulomb-(SR)	10 Coul.-recip.	11 Potential	12 Box-X
13 Box-Y	14 Box-Z	15 Volume	16 Density
17 Coul-SR:Protein-Protein	18 LJ-SR:Protein-Protein		
19 Coul-14:Protein-Protein	20 LJ-14:Protein-Protein		
21 Coul-SR:Protein-SOL	22 LJ-SR:Protein-SOL		
23 Coul-14:Protein-SOL	24 LJ-14:Protein-SOL		
25 Coul-SR:Protein-BEN	**26 LJ-SR:Protein-BEN**		
27 Coul-14:Protein-BEN	**28 LJ-14:Protein-BEN**		
29 Coul-SR:Protein-rest	30 LJ-SR:Protein-rest		
31 Coul-14:Protein-rest	32 LJ-14:Protein-rest		
33 Coul-SR:SOL-SOL	34 LJ-SR:SOL-SOL		
35 Coul-14:SOL-SOL	36 LJ-14:SOL-SOL		
37 Coul-SR:SOL-BEN	38 LJ-SR:SOL-BEN		
39 Coul-14:SOL-BEN	40 LJ-14:SOL-BEN		
41 Coul-SR:SOL-rest	42 LJ-SR:SOL-rest		
43 Coul-14:SOL-rest	44 LJ-14:SOL-rest		
45 Coul-SR:BEN-BEN	46 LJ-SR:BEN-BEN		
47 Coul-14:BEN-BEN	48 LJ-14:BEN-BEN		
49 Coul-SR:BEN-rest	50 LJ-SR:BEN-rest		
51 Coul-14:BEN-rest	52 LJ-14:BEN-rest		
53 Coul-SR:rest-rest	54 LJ-SR:rest-rest		
55 Coul-14:rest-rest	56 LJ-14:rest-rest		

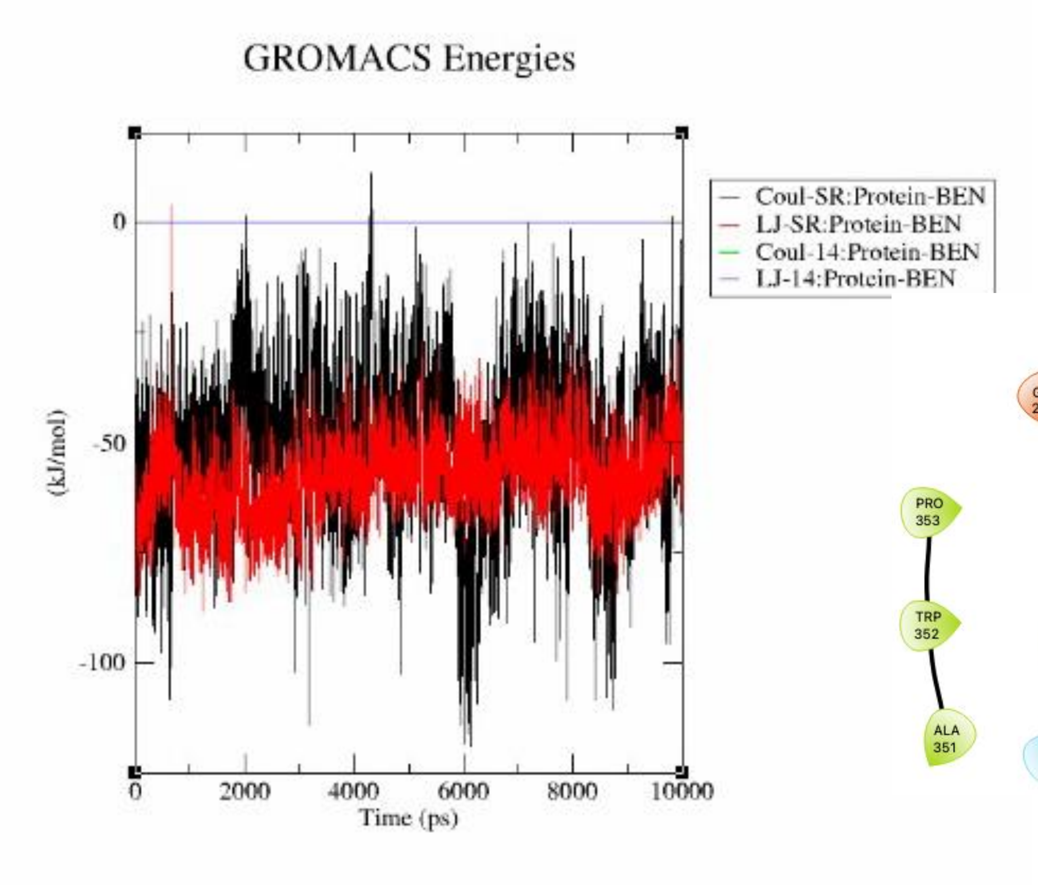


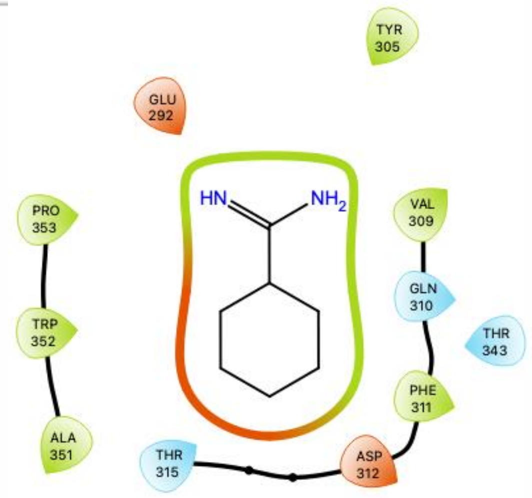
Command line:

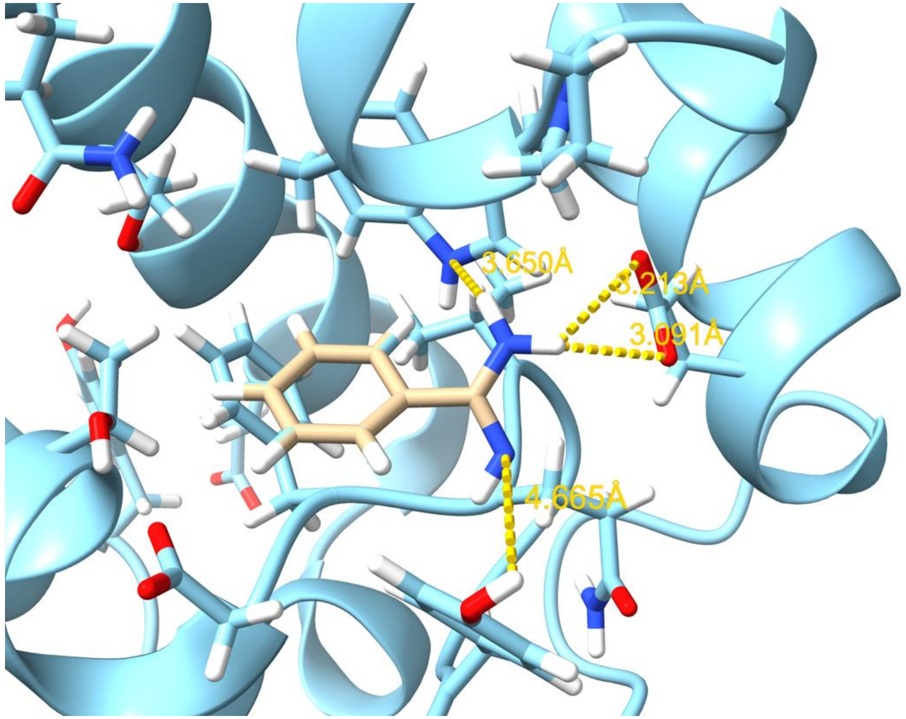
gmx energy -f energy_NPT_rerun.edr –sum (opt)

(selecionar termos não ligados)

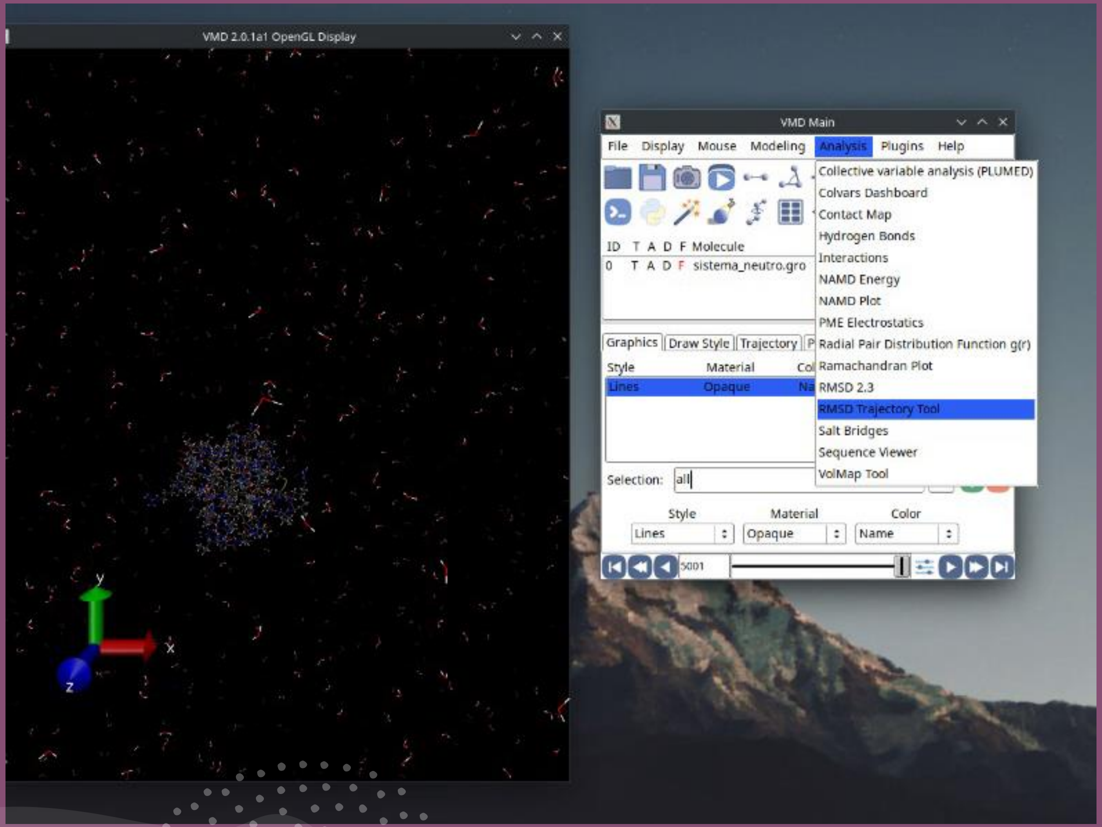
Análises: Energia de interação proteína-ligante





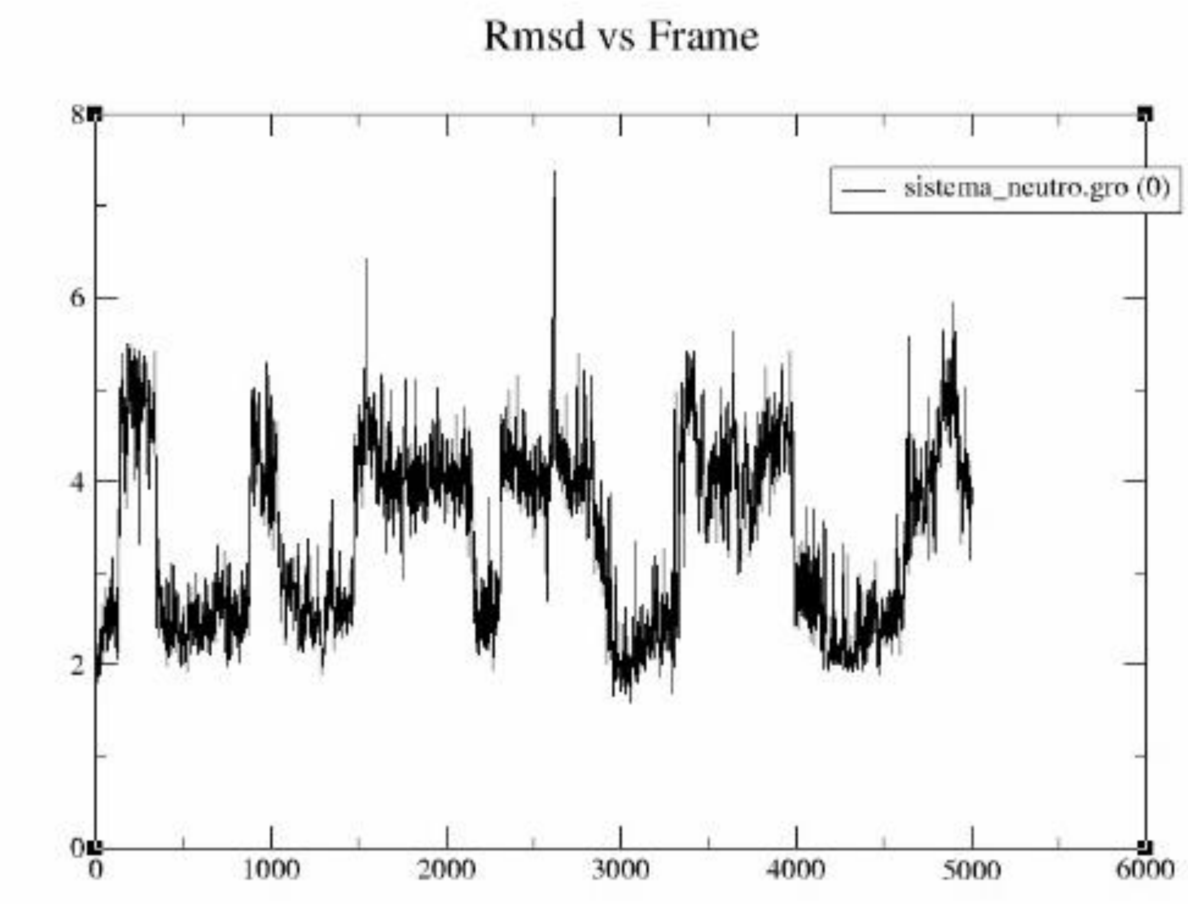


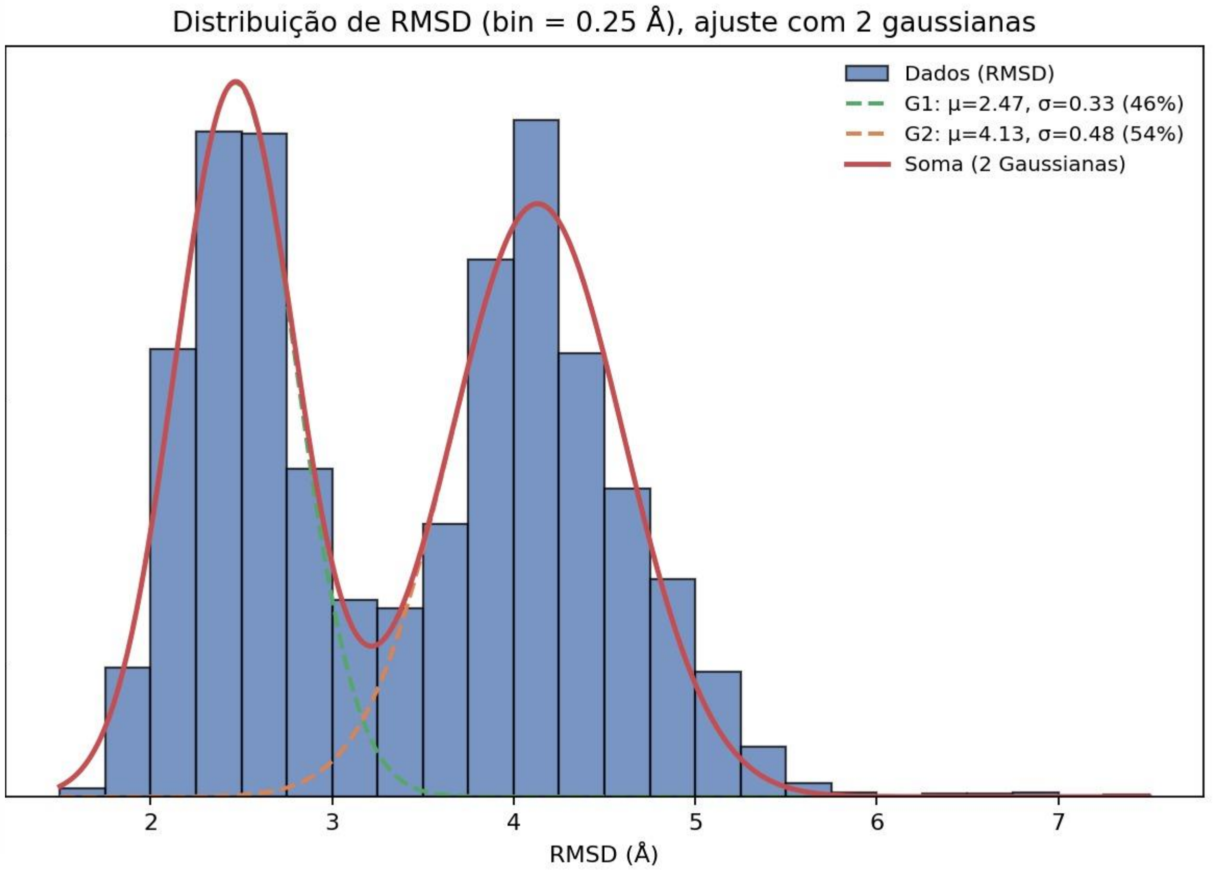
Análises: Raiz do Desvio Quadrático Médio (RMSD)



- Utilizar o VMD
  - Carregar a trajetória da proteína com o ligante (vmd ../EM/em.tpr md_NPT_10ns.xtc)
  - Ir para Analisis/RMSD Trajectory Tool
  - Alinhar a estrutura da proteina à referência cristalina. (Selecionar protein)
  - Calcular RMSD para o ligante (Selecionar resname BEN)
  - Salvar o arquivo para Xmgrace

Análises: Raiz do Desvio Quadrático Médio (RMSD) - Benzamidina



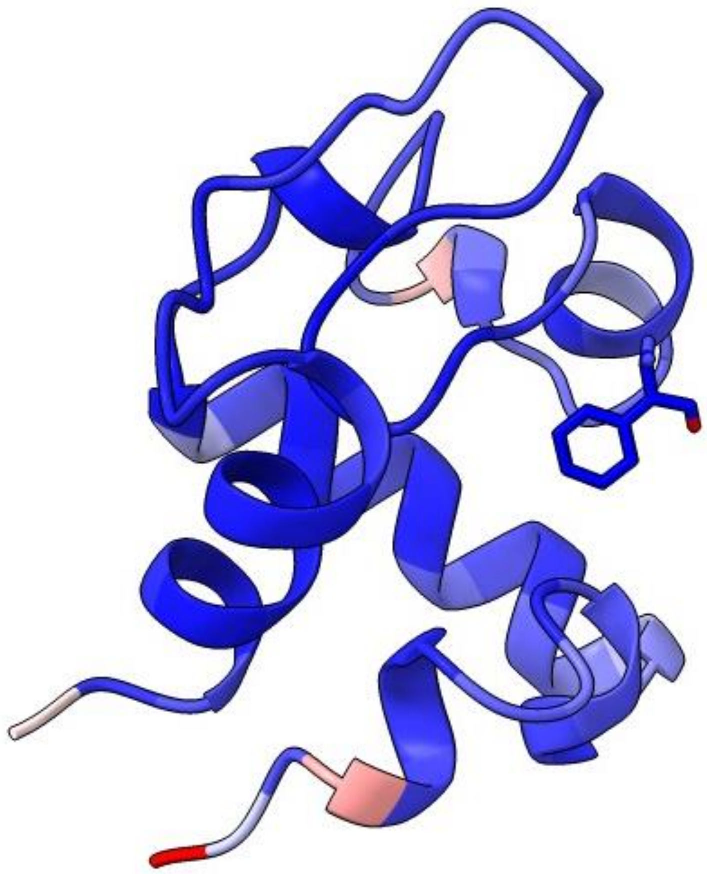


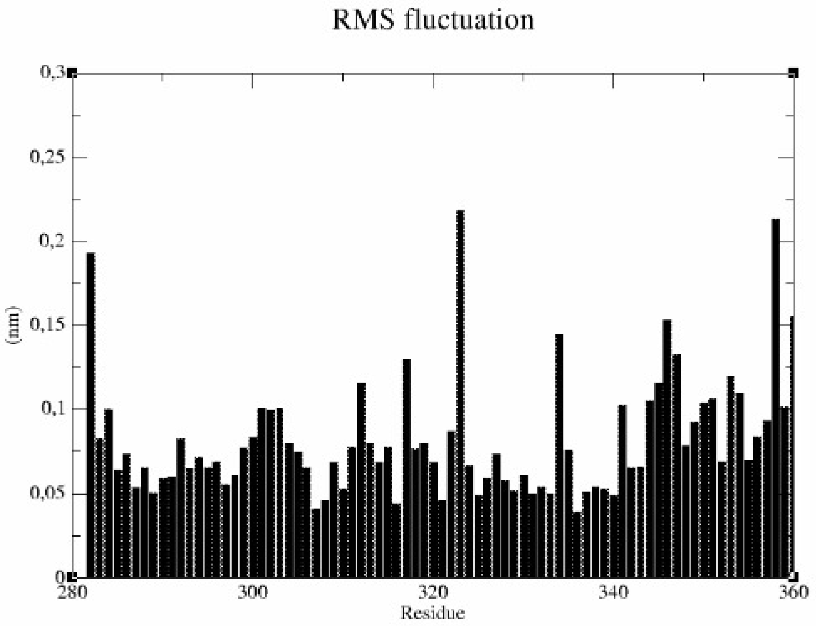
Análises: Flutuações (RMSF)

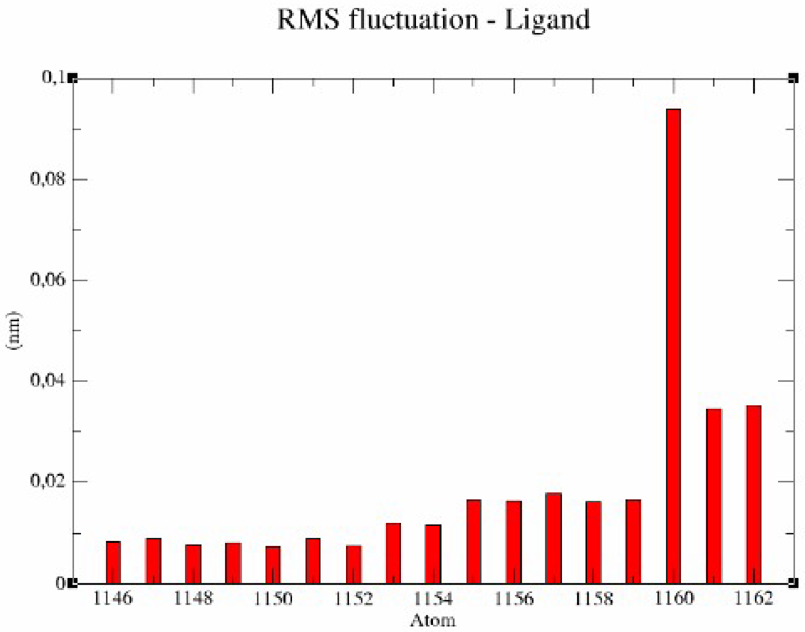
Command line:

gmx rmsf -s **../EM/em.tpr** -f md_NPT_rerun.trr -res -oq

- Considerar primeiro a proteina
- Considerar depois o ligante





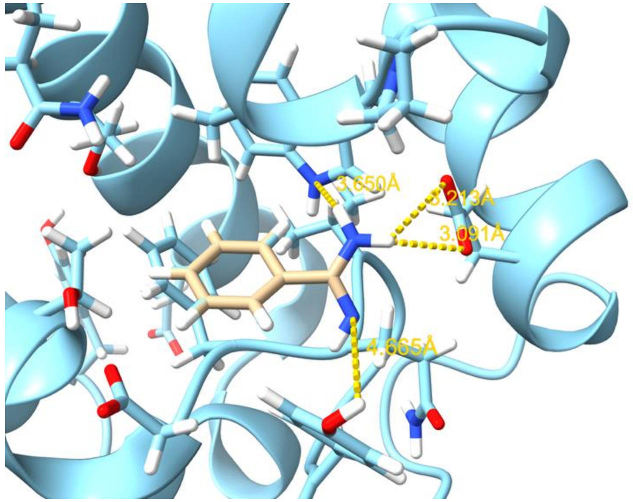


H-Bonds

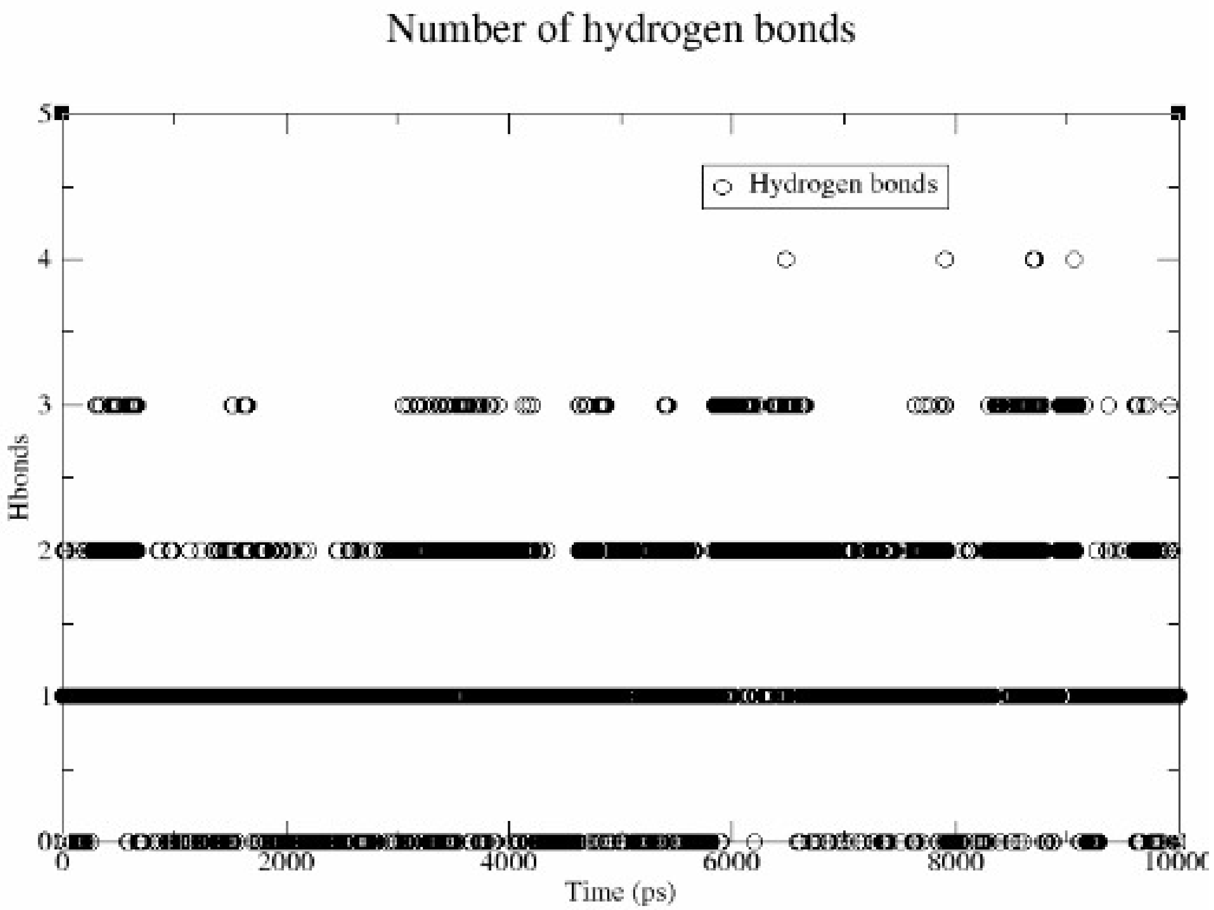
```
Reading file md_NPT_10ns.tpr, VERSION 2024.5 (single precision)
Reading file md_NPT_10ns.tpr, VERSION 2024.5 (single precision)
Available static index groups:
 Group  0 "System" (14277 atoms)
 Group  1 "Protein" (1145 atoms)
 Group  2 "Protein-H" (596 atoms)
 Group  3 "C-alpha" (80 atoms)
 Group  4 "Backbone" (240 atoms)
 Group  5 "MainChain" (321 atoms)
 Group  6 "MainChain+Cb" (387 atoms)
 Group  7 "MainChain+H" (401 atoms)
 Group  8 "SideChain" (744 atoms)
 Group  9 "SideChain-H" (275 atoms)
 Group 10 "Prot-Masses" (1145 atoms)
 Group 11 "non-Protein" (13132 atoms)
 Group 12 "Other" (17 atoms)
 Group 13 "BEN" (17 atoms)
 Group 14 "NA" (2 atoms)
 Group 15 "Water" (13113 atoms)
 Group 16 "SOL" (13113 atoms)
 Group 17 "non-Water" (1164 atoms)
 Group 18 "Ion" (2 atoms)
 Group 19 "Water_and_ions" (13115 atoms)
Specify a selection for option 'r'
(Reference selection, relative to which the search for hydrogen bonds in target selection will develop.):
(one per line, <enter> for status/groups, 'help' for help)
```

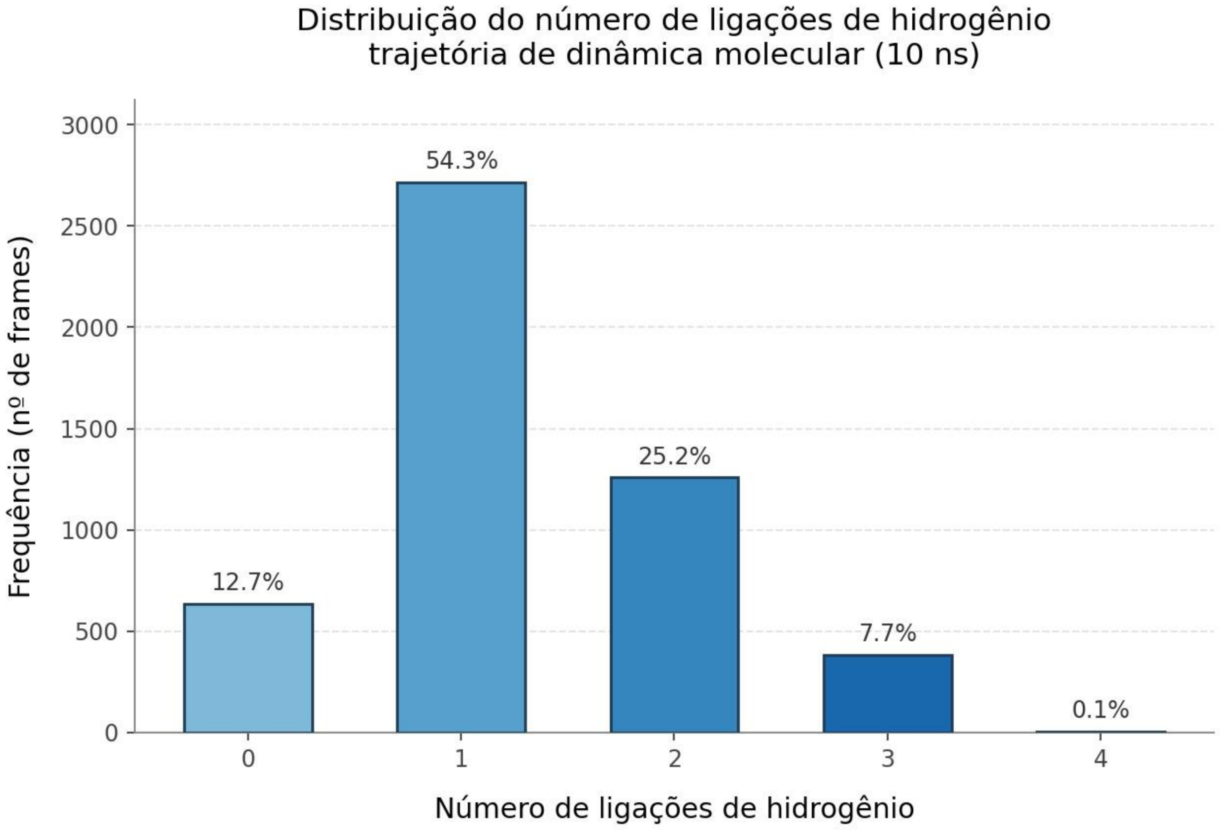
Command line:

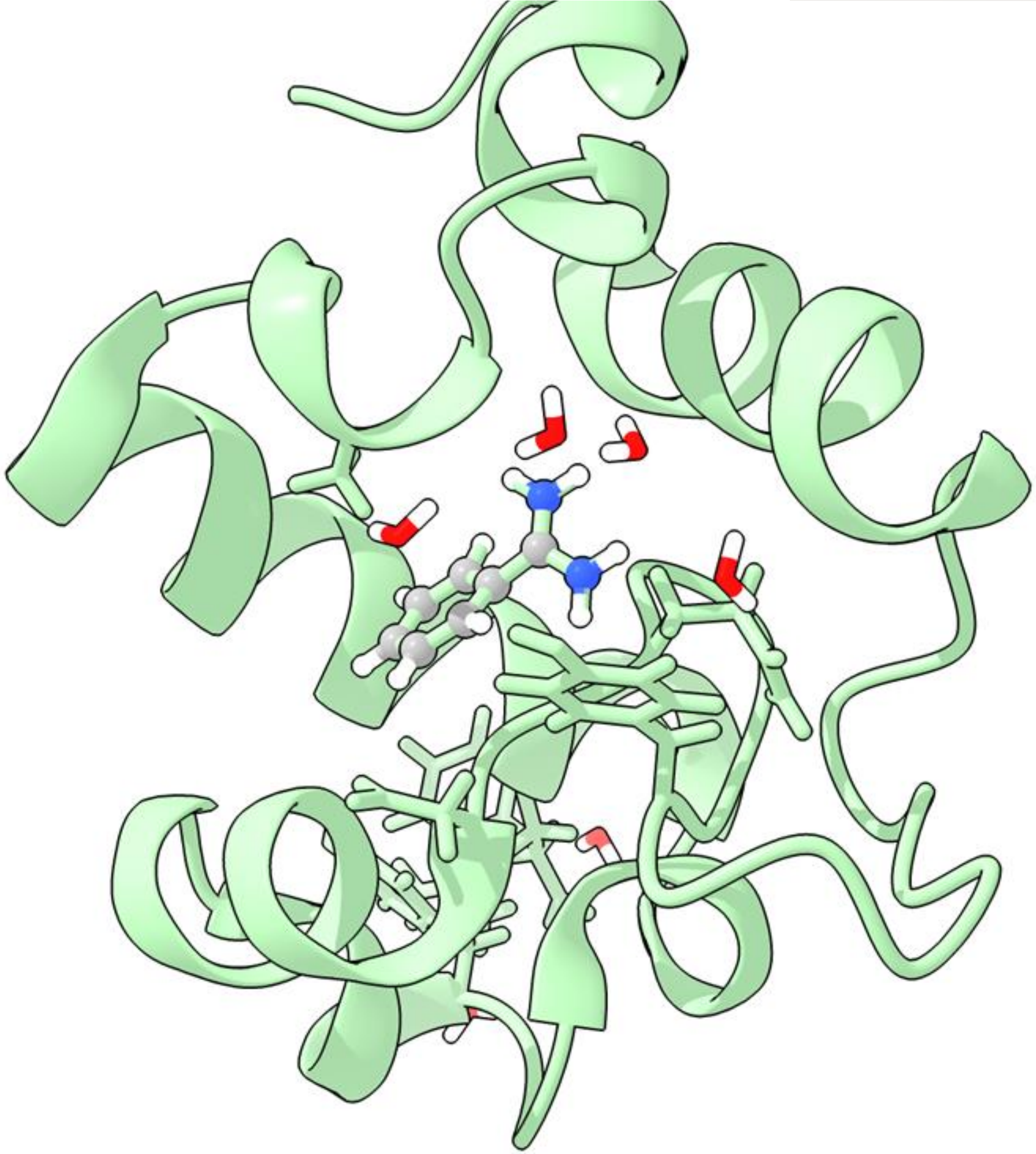
gmx hbond -s md_NPT_10ns.tpr -f md_NPT_rerun.trr -num hbonds



Análises: Ligações Hidrogênio Proteína - Ligante







FIM